

VENKATA SAI SANATH NARAYAN KAVATOORU

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Experience

Rule4

Boulder, CO

Software Engineer Intern | NextJS, FastAPI, PostgreSQL, AWS, GCP

Oct 2025 – Jan 2026

- Led development of a full-stack resource scheduling platform from scratch, enabling preference-based allocation with points system, waitlisting, and auto-reallocation for **50+ premium resources**.
- Connected the **Google Calendar API** for two-way sync, enabling real-time availability updates and reducing scheduling conflicts by **40%**.
- Designed **RESTful APIs** with **FastAPI** and **PostgreSQL**, implementing role-based access control (RBAC), audit logging, and OWASP security best practices.
- Launched a production-ready application on **AWS RDS** and **S3** with **GCP** infrastructure, delivering a scalable enterprise solution adopted by the company.

AirAsia MOVE

Bangalore, Karnataka

Software Engineer | Swift, UIKit, MVVM, Combine

Feb 2022 – Jul 2024

- Shipped a social profile screen in **UIKit** and refreshed chat theming (fonts, wallpapers, colors), adopted by **60%** of monthly active users.
- Refactored chat UI into **MVVM** and consolidated **15** legacy screens into reusable components, reducing new UI development time by **35%**.
- Collaborated with backend to migrate chat from a third-party provider to an in-house stack, saving **\$12K/month** and improving product control.
- Transitioned Promise-based async flows to **Combine** pipelines to strengthen cancellation/retry handling and improve UI/network state consistency.
- Authored **50+ XCUITest** UI tests running on every build and nightly with CI reports, catching regressions earlier and speeding **TestFlight** releases.
- Automated a Ruby CI pipeline integrated with **Git** to publish real-time pass/fail dashboards and artifacts, cutting regression triage time by **50%**.
- Optimized a streaming GIF renderer (replacing SDWebImage GIF handling), reducing peak memory per chat thread by **65%** via Xcode Instruments.

Projects

Medical Document Q&A Assistant (RAG) | Python, FastAPI, Next.js, LangChain, OpenAI

[Live Demo](#)

- Created a **full-stack RAG assistant** (Next.js, FastAPI) enabling grounded Q&A over PDFs using **LangChain**, **OpenAI**, and **vector search** (chunking, top-k retrieval) to minimize hallucinations.
- Established **REST APIs** integrated with **AWS S3** to persist **vector stores**, enabling seamless pulling and pushing of embedding updates for stateful sessions.

AI Learning Assistant | Python, Streamlit, LangGraph, LangChain, ChromaDB, RAG, STT/TTS

[GitHub](#)

- Orchestrated an adaptive learning platform using **Streamlit** and **LangGraph**, coordinating a **finetuned LLM** and **multi-modal RAG pipeline** (PDF/Images) with **ChromaDB**.
- Served voice-first quizzes and mock interviews by integrating **STT/TTS** and hosting inference on **GPU** infrastructure for low-latency responses.

URL Shortener | ReactJS, Spring Boot, PostgreSQL, Redis, Docker, Kubernetes, Google Cloud

[Live Demo](#)

- Engineered a high-performance URL shortener with **Spring Boot** and **PostgreSQL** on **Google Cloud**, implementing **Redis caching** to handle high-traffic redirects with minimal latency.
- Developed a **ReactJS** dashboard for analytics and link management, and provisioned containerized microservices using **Docker** and **Kubernetes** on **Google Cloud**.

AudioTranscriptor | Swift, AVAudioEngine, AWS, OpenAPI

[GitHub](#)

- Crafted a native iOS app in **Swift** handling lengthy recordings (2+ hours) by chunking audio with **AVAudioEngine** and implementing a robust, resumable upload pipeline to **AWS**.
- Achieved a seamless transcript playback experience with **<100ms sync latency**, optimizing navigation for long audio files using efficient buffering techniques.

Skills

Languages: Python, Swift, C++, JavaScript, TypeScript, SQL, Bash

Frontend: React, Next.js, Tailwind CSS, HTML/CSS, Streamlit

Mobile: UIKit, SwiftUI, Combine, XCUITest

Backend: Spring Boot, FastAPI, Django, Node.js, NestJS, REST

Cloud: AWS (EC2, RDS, S3), Google Cloud, Docker, Kubernetes, CI/CD (GitHub Actions)

Data: PostgreSQL, MySQL, Redis, ChromaDB, MongoDB

AI: LangChain, LangGraph, RAG, Fine-tuning, STT/TTS, Embeddings

Education

University of Colorado Boulder

Aug 2024 - May 2026

Master of Science, Computer Science

CGPA: 3.84/4